UAV Perception Evaluation: Resolution, Slicing and Selection Reliability

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Technical report | Controlled evaluation | Not peer reviewed

Code, aggregate results and reproducibility material.

Abstract

This study evaluates four established UAV object-detection inference configurations using three trained YOLO11n checkpoints, 548 development images and a frozen confirmation on 1,610 VisDrone test-dev images. Full-frame 1280 leads all 18 development checkpoint-by-scoring-convention comparisons and all three primary-metric test-dev comparisons. Test-dev AP is 19.260-19.486%, a same-checkpoint cross-split decrease of 3.712-4.421 percentage points from development. A known AP-integration change reverses the close slicing/full-960 development ranking for two checkpoints without changing predictions. Retrospective uniform-subset selection recovers the full-development-pool winner in 69-85% of 64-image draws and 93-100% of 256-image draws; one 16-image draw loses 4.455 AP points. The findings support larger full-frame controls and explicit evaluation objectives, not a new detector or universal annotation budget. Test-dev labels are public, scene independence remains unknown, and scorer controls and label-budget results remain development-specific. Full public reproduction and publication novelty are unestablished.

1. Motivation and position relative to prior work

Small aerial objects can lose useful detail when an entire image is resized. Higher full-frame resolution and sliced inference are established responses, but they change computation, object scale, context and duplicate handling. VisDrone provides a relevant drone-image benchmark [1]. SAHI already demonstrates slicing-assisted inference and fine-tuning for small-object detection [3]; neither slicing nor applying it to aerial images is a contribution of this study. YOLO11n supplies the existing detector [2]. The fixed-slicing implementation here is an experimental control, not an exact SAHI-author reproduction.

The questions are deliberately narrower: with the same trained weights, which of four fixed configurations has the strongest development AP; does the primary ordering transfer to a separate official split without retuning; how stable are close development rankings to declared evaluator choices; and how often does partial-image evaluation select the full-development-pool winner? Existing active-testing work already addresses economical performance estimation, including AP-related tasks [6]. Work distinguishing estimation from model selection also motivates measuring selection regret rather than only score-estimation error [7]. This report supplies a controlled empirical investigation of these questions, not a new acquisition algorithm or estimator.

The contribution is the combination of retained evidence, controlled comparisons and explicit limits: three training checkpoints, shared-weight inference comparisons, version-specific development scoring controls, frozen test-dev confirmation, a finite-pool development label-budget diagnostic and a scoped portable execution path. No claim of method novelty or peer-reviewed acceptance follows from that combination.

2. Data, training and evaluation protocol

2.1 Data and task definition

The study uses 6,471 official VisDrone2019-DET training images and all 548 validation images. The curated ten-class task contains 343,200 training targets and 38,759 validation targets. Native score-1 categories 1-10 map to indices 0-9. Three exact known malformed training rows are excluded and retained in exception records; four duplicate float32 training labels are removed. Originals remain unchanged. Excluded and ignored annotations are retained separately, but ignored pixels are not masked in the converted training loss or primary metric. Consequently, these converted-task scores are not official native VisDrone challenge scores.

Validation was used for checkpoint selection and subsequent analysis. It is exposed development data, not untouched final-test evidence. Repeating it across checkpoints does not multiply the number of independent images.

The subsequent official test-dev split contains 1,610 image/annotation pairs and 77,547 native rows. The same declared task conversion yields 75,102 scored targets and 2,445 excluded rows, with zero new malformed exceptions, clipping or duplicate-label removals. All converted labels, exclusion records and copied-image identities were checked against the originals. Test-dev ignored pixels are likewise not masked in the primary metric.

2.2 Training and inference controls

Three YOLO11n models start from the same pretrained checkpoint and complete 150 epochs at training image size 640, batch 16. The declared training settings use SGD, initial learning rate 0.01, cosine decay, momentum 0.937, weight decay 0.0005, three warm-up epochs, mosaic probability 1.0 and mosaic closure for the final 15 epochs. Mixed precision is used during training. Best development AP50-95 selects epochs 95, 103 and 114 for seeds 0, fresh 1 and 2. An incomplete seed-1 attempt was stopped for an operational failure and retained separately; the completed fresh replacement starts again from the original pretrained weights with unchanged settings.

Each checkpoint is evaluated at full-frame input sizes 640, 960 and 1280, and with fixed 640-pixel tiles, nominal 20% overlap, a global 640 pass and CPU class-aware merging. All four configurations share the same checkpoint within a seed. They are not separately trained or optimized at their respective inference resolutions. Inference uses FP32, batch 1, square padding, confidence floor 0.001, one highest-scoring class per proposal, detector and merge NMS IoU 0.7, and 1,000-output caps. Predictions are retained before the scoring controls.

2.3 Metrics and computation

Primary AP50-95 is the mean class AP over ten classes and ten IoU thresholds from 0.50 to 0.95, using the pinned Ultralytics 8.3.253 implementation. Detections are matched within images and pooled by class for scoring. AP is not mean per-image AP. Evaluator version, matching, cap, class aggregation and precision-recall integration policy are part of the reported metric.

On development, the four configurations consume relative input-pixel totals 1.000, 2.250, 4.000 and 6.186. Fixed slicing requires 3,390 model passes versus 548 for each full-frame configuration. These are deterministic workload counts, not energy or hardware-independent runtime estimates. Desktop timings drifted across repeated sweeps, so no stable speedup or onboard-flight latency is claimed.

3. Main development results and evaluator sensitivity

Inference configuration	Seed 0 AP (%)	Fresh seed 1 AP (%)	Seed 2 AP (%)
Full 640	18.9660	18.5768	18.7431
Full 960	22.8522	22.6016	22.7420
Full 1280	23.6807	23.4520	23.1983
Fixed slices + global	22.4980	21.9473	22.3718

Table 1. Pinned converted-task AP50-95, cap 1,000. Each column uses one checkpoint across all four configurations. The same 548 development images are used throughout.

Full 1280 is highest among these four configurations under each of six named scoring conventions for all three checkpoints: 18 descriptive comparisons. The conventions reuse saved detections and therefore are controls, not 18 independent training experiments. They do not establish universal superiority or a statistical population claim.

The close full-960/slicing comparison is more sensitive:

Scoring convention	Seed 0 ΔAPpp	Fresh seed 1 ΔAPpp	Seed 2 ΔAPpp
Pinned converted, cap 1,000	-0.3542	-0.6544	-0.3702
Pinned converted, cap 500	-0.3149	-0.6274	-0.3647
Legacy class macro, cap 500	+0.0962	-0.1821	+0.1475
Legacy source aggregate, cap 500	+0.2604	+0.0281	+0.2423
Published AP boundary, cap 1,000	+0.2062	-0.0459	+0.2415
Published AP boundary, cap 500	+0.1178	-0.1489	+0.1550

Table 2. Slicing minus full-960 AP, in percentage points; positive favours slicing. The known published AP-boundary change alone reverses the cap-1,000 ordering in two of three checkpoints [4]. Legacy aggregation introduces additional policy differences and is not an isolated boundary-only contrast.

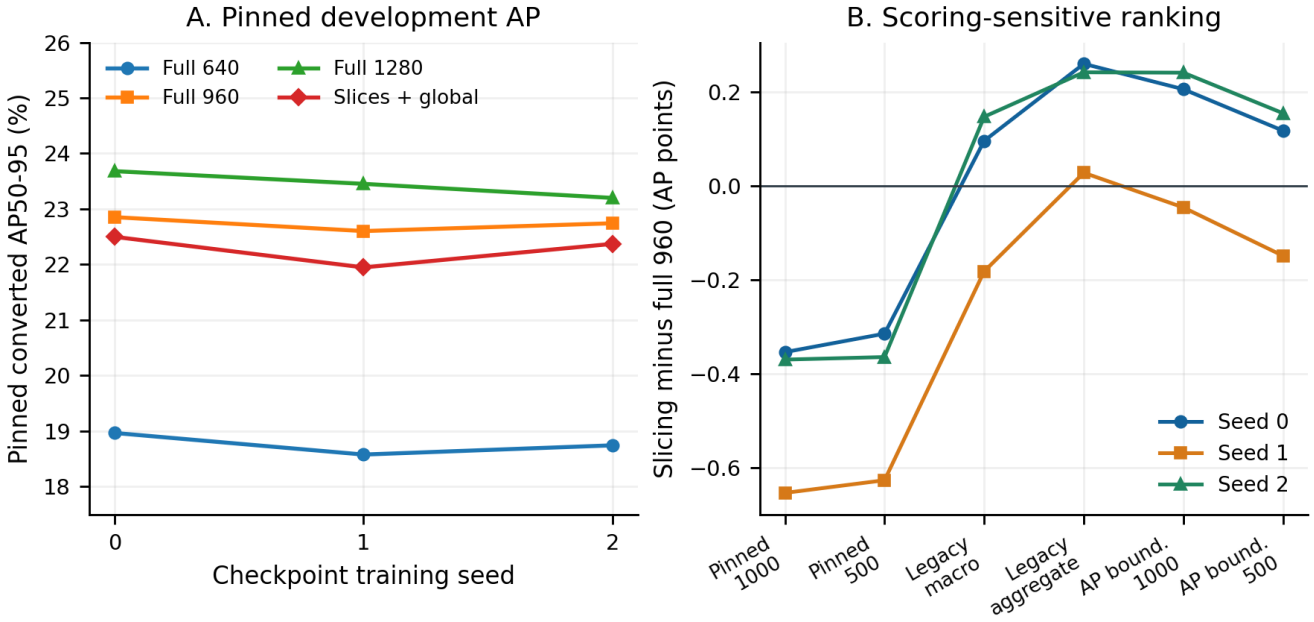


Figure 1. Left: pinned converted AP across three checkpoints. Right: slicing-minus-full 960 under six conventions. Positive values favour slicing. The same retained detections support the scorer contrasts; these are not new training experiments.

4. Reference checks and an operating-point trade-off

A separate development-only current-matcher control uses the reviewed function at upstream commit `abc2066dd6bd87f6806f73a170cf8f764ca2ebbe`, on seeds 0 and 1 only [5]. It changes the matching function while retaining saved boxes, confidence order and other declared policies. Across the checked scoring/retention combinations, its AP50-95 shifts are +0.008224 to +0.022647 percentage points. These are evaluator changes on identical outputs, not detector improvements. Full 1280 remains highest. Related upstream work already existed; no new matcher bug or fix is claimed, and the installed framework was not upgraded.

For a different development-only reference check, the legacy scoring port agrees with unchanged captured publisher functions under GNU Octave 11.3.0 for all four seed-0 settings and all 548 images. The largest aggregate difference across seven metrics is 2.487×10^{-13} percentage points. This is exact-input/reference agreement within the checked scope, not exhaustive MATLAB equivalence or an official challenge submission. Seeds 1 and 2 have ported legacy scores without this same exact-publisher execution check.

Seed-0 error analysis adds a distinction between pooled AP and an operating point. At confidence at least 0.25 and IoU at least 0.50, slicing has 61.27% matched-target recall versus full-1280's 56.33%, but precision falls from 62.35% to

44.22%. Slicing adds 1,915 net matched targets and 16,774 unmatched predictions relative to full 1280. Scores are not calibrated across configurations, so the common numerical confidence threshold is not an equal-precision comparison.

Seed-0 setting	Recall: area <256 px²	Recall: 256-<1,024	Recall: 1,024-<9,216	Recall: ≥9,216	Precision
Full 1280	35.16%	57.31%	75.33%	82.12%	62.35%
Slices + global	38.24%	62.94%	81.13%	89.51%	44.22%

Table 3. Seed-0 recall by original integer target area, confidence ≥0.25 and IoU ≥0.50. These are declared diagnostic bins, not official AP-small categories. Other checkpoints were not included in this size analysis. An unmatched output under the converted task is not automatically a visually nonexistent object, because excluded native annotations have a different policy.

5. Retrospective label-budget selection

The diagnostic asks whether pooled AP on a uniformly selected image subset recovers the full-pool best configuration. For checkpoint s, let Q_s(k) be full-pool AP of configuration k and let the subset-selected configuration be k̂. Reported regret is

R_s(k̂) = max_k Q_s(k) - Q_s(k̂), in AP percentage points.

This is the quality loss from a selection decision. It is not the difference between subset AP and full-pool AP. A previously informed fixed full-1280 choice has zero retrospective regret here; it is an explicit comparator, not a discovery made without labels.

The internally frozen protocol uses 100 deterministic PCG64 permutations, nested image budgets 16/32/64/128/256/548, the same masks for every configuration/checkpoint and the primary pinned converted metric. Queried images return to canonical source order before pooled scoring, preserving equal-score ordering. Selection receives only revealed statistics; the full-pool oracle evaluates the chosen configuration afterwards. An absent GT class would trigger abstention, and exact ties use a declared candidate order. All observed subsets contain all ten classes, so no decision abstains.

Images	Exact winner /100: seed 0 / 1 / 2	Mean regret APpp: seed 0 / 1 / 2
16	66 / 53 / 48	0.3419 / 0.5305 / 0.3772
32	69 / 72 / 58	0.3064 / 0.3166 / 0.2842
64	84 / 85 / 69	0.1503 / 0.1537 / 0.1859
128	97 / 96 / 82	0.0284 / 0.0340 / 0.0932
256	100 / 98 / 93	0.0000 / 0.0170 / 0.0319
548	100 / 100 / 100	0.0000 / 0.0000 / 0.0000

Table 4. Selection outcomes for the exposed finite development pool. Full 548 is one unique subset per checkpoint; its 100 rows are repeated full-pool decisions, not independent trials.

One seed-2 draw at 16 images selects full 640 and incurs **4.455192 APpp** regret. That group's mean is 0.377232 and its 90th percentile 0.826507; both hide this maximum. Even at 256 images, exact-winner recovery is 93/100 for seed 2. No independently justified practical tolerance or failure probability was specified, so no budget is certified sufficient.



Figure 2. Retrospective exact-winner recovery (left) and maximum observed full-pool regret (right), not mean regret. All curves use 100 nested query orders on 548 exposed images. Full 548 is one unique subset per checkpoint; the rare 4.455-point loss is retained.

6. Frozen test-dev confirmation

6.1 Prospective choices and execution scope

An internal plan and scope clarification were fixed before test-dev acquisition, label inspection, predictions or scoring. Earlier development findings were known, so this is prospective split confirmation within the investigation, not external preregistration. All three already selected checkpoints and all four inference configurations were retained. There was no retraining, checkpoint reselection, threshold/resolution/NMS/tile tuning or outcome-based configuration change on test-dev.

All 1,610 images were processed under the same pinned converted ten-class primary metric and inference settings. The complete runs retain 19,320 image/configuration records and 48,834 detector passes, with warm-ups recorded separately. These are repeated configurations and checkpoints on the same 1,610 images, not 19,320 independent test images. The three run consoles contain no flagged warning/error/NMS-time-limit lines. This is one inference per image/configuration/checkpoint, not a repeated timing or determinism experiment.

A separate CPU execution recomputed all twelve complete primary summaries exactly from saved predictions and labels without using the original summaries as scoring inputs. It executes the **same pinned matching/AP algorithm**: agreement checks execution and retained-output integrity, not an independently implemented evaluator or external validation. The development study's six scoring conventions, seed-0 publisher-reference check, two-seed current-matcher control, size analysis and label-budget experiment were not repeated on test-dev.

6.2 Complete primary results

Inference configuration	Seed 0 test-dev AP (%)	Seed 1 test-dev AP (%)	Seed 2 test-dev AP (%)
Full 640	14.7877	14.9810	14.8263
Full 960	18.5950	18.8040	18.6282
Full 1280	19.2601	19.3143	19.4858
Fixed slices + global	18.3658	18.4196	18.3992

Table 5. Test-dev AP50-95 under the unchanged primary converted task. This is not native VisDrone challenge scoring; no six-convention test-dev result is claimed.

Checkpoint	Full 1280 minus 960 (APpp)	Full 1280 minus slices (APpp)	Full 1280 test-dev minus development (APpp)
Seed 0	+0.6651	+0.8943	-4.4206
Seed 1	+0.5103	+0.8947	-4.1377
Seed 2	+0.8576	+1.0866	-3.7125

Table 6. The first two contrasts were specified in the internal plan. The final column is the same-checkpoint descriptive cross-split change for full 1280, not a causal estimate of overfitting or distribution shift.

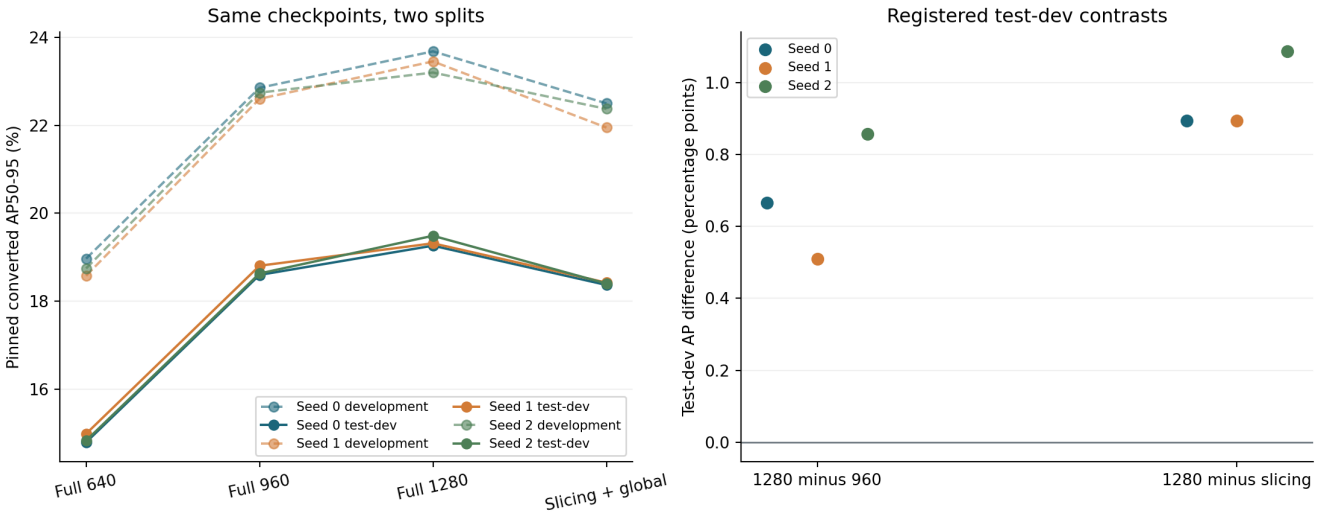


Figure 3. Development and test-dev AP for the same three selected checkpoints, with six registered test-dev contrasts. Solid and dashed lines distinguish the splits. Checkpoint points are not independent scenes, and no population confidence or significance claim is shown.

Full 1280 remains the highest-AP setting in all three test-dev comparisons. Its advantage is 0.510-0.858 points over full 960 and 0.894-1.087 over this slicing recipe. This confirms the primary development ordering on the separate official split, while absolute AP is substantially lower. Full-1280 AP falls from 23.198-23.681% on development to 19.260-19.486% on test-dev; each checkpoint loses 3.712-4.421 points. Cross-split AP alone does not identify why: selection optimism, scene composition, annotation characteristics and other factors were not isolated experimentally.

6.3 Cost and what the confirmation does not establish

Each full setting makes 1,610 detector passes per checkpoint, versus 11,448 for slicing plus global. Slicing consumes 7.11056 times the full-640 square-input pixels, or 1.77764 times full 1280. The different development ratio reflects the new images' dimensions. Thus full 1280 has higher primary AP and lower input-pixel count than this slicing recipe on test-dev, within these fixed controls. Pixels are not FLOPs, energy or a general efficiency frontier.

Full-1280 diagnostic median times vary from 16.17 to 31.39 ms across sequential checkpoints; slicing varies from 62.00 to 137.61 ms. Such drift prevents a stable speedup or onboard-deadline claim. No favourable timing sweep is selected as the headline.

The public test-dev labels were held out from this investigation's development decisions. Exact-byte disjointness checks found no cross-split image groups, but this does not exclude near duplicates, shared scenes or pretraining exposure. No verified scene map was available in the supplied archive. Consequently, this is neither a blind challenge test nor verified scene-independent or out-of-distribution performance. The confirmation strengthens one bounded comparison; it does not externally validate every development diagnostic or establish a new method.

7. Interpretation, reproducibility and next decision

The label-budget grid still concerns the exposed 548-image development pool. Its 1,800 decisions are not trained-model replications. Independent recomputation of 48 AP scores agrees exactly with retained results. Sixteen-image subsets contain 754-1,592 GT boxes, median 1,166: image count is not annotation time or constant label effort. Existing-label retrospective selection establishes neither actual annotation saving nor a new active method. Test-dev confirmation of the full-pool ranking does not validate these selection curves on a new pool.

The implementation retains source/configuration identities, checkpoint hashes, data audits and saved predictions. A separately versioned preparer recreates the curated train/validation task from raw files and verifies semantic content as well as actual record bytes. A prospective execution path connects preparation, training, completion review, inference and rescoring, while preserving exact-byte resume requirements. Real full-data preflight and one-image/four-configuration CPU inference/rescoring verify bounded plumbing. The development software package has 121 passing synthetic CPU controls. Counts of checks do not prove complete correctness or detector quality. Test-dev uses a separate split adapter and full-data execution, not a relabelling of the training/validation preparation.

The companion software supplies source, configurations, synthetic controls, aggregate tables and rendering; it does not distribute images, labels, weights or raw predictions. Its native/reference/current-matcher results are aggregate-only, and real full-grid historical-input orchestration and sufficient-statistic caches are outside it. The test-dev extension supplies protocol and numerical aggregates; its internal full-split execution adapter and raw predictions are not included. These aggregates do not reproduce raw AP or detector inference. Read the exact software release's scope. Clean-environment installation, fresh prospective 150-epoch training, a full train/validation portable rerun and complete public AP reproduction remain unverified. The original checkpoints retain their actual training provenance.

The principal limitations remain one detector family, three development-selected checkpoints, no inference-resolution-specific retraining, converted ignore semantics, partial cross-seed/split scope for auxiliary controls, timing drift and unknown scene-level dependence. No confidence interval, statistical population significance, robot integration, flight safety or state-of-the-art claim is established.

The completed confirmation does not justify adding a learned selector for appearance. A stronger research question should start from a practical decision endpoint, acceptable loss and acquisition/deployment cost, then establish a meaningful failure of strong fixed and simple sampling controls. Scene-aware evaluation requires verified grouping information; an AP-aware selection extension requires its own estimand, assumptions and close prior-work comparison. These are next questions, not completed methods or proven gaps.

Conclusion. Full-frame 1280 is the strongest primary AP choice among these four configurations on both development and frozen test-dev evaluation. Close development alternatives remain scorer-sensitive, and small uniform subsets can produce costly selection errors. The 3.712-4.421-point cross-split drop also shows why stable ranking is not the same as stable absolute performance. The defensible result is a documented controlled investigation with a completed bounded confirmation. New-method novelty and peer-reviewed contribution remain unestablished.

References

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Primary-source availability checked 18 October 2026; experimental evidence comprises the retained development records, frozen test-dev execution and same-algorithm rescoring records, and their aggregate exports.